DURGA PRASAD KOTA

AI/ML Engineer | Python Backend Developer

7286945898 durgaprasad04289@gmail.com Portfolio LinkedIn GitHub Eluru, India

PROFESSIONAL SUMMARY

AI/ML Engineer and Python Backend Developer experienced in **Python, Django, FastAPI, and REST APIs**. Skilled in building scalable backend systems, ML-powered applications, and full-stack solutions using **React.js** and **PostgreSQL**. Experienced in **Agentic AI, LLMs, LangChain, LangGraph, RAG, and MCP**, with expertise in integrating vector databases, embeddings, and AI agents into production-ready applications.

EDUCATIONS

Ramachandra College of Engineering | Eluru, Andhra Pradesh — 2024 - 2028
B.Tech in Artificial Intelligence and Machine Learning Engineering — *CGPA : 8.9*

Venkateswara Junior College | Jangareddygudem, Andhra Pradesh — 2022 - 2024
Intermediate in Maths, Physics and Chemistry — *CGPA : 95.6%*

Zilla Parishad High School | East Edavalli , Andhra Pradesh — 2020 - 2022
Central Board of Secondary Education — *CGPA : 90.0%*

PROJECTS

HireLense – AI-Powered Resume Intelligence & ATS Evaluation Platform | Github Website

- Architected an **AI-powered resume intelligence platform** using FastAPI, Groq API, LangChain, and RAG pipelines for contextual resume retrieval and structured information extraction.
- Implemented secure authentication with **JWT access/refresh tokens**, Google OAuth 2.0, Redis-backed OTP verification, and SlowAPI rate limiting.
- Engineered **PostgreSQL and Redis caching infrastructure** using SQLAlchemy and Upstash Redis, optimizing API performance and reducing response latency by **80%**.
- Developed a responsive SaaS dashboard with **React.js, Vite, and Chart.js**, featuring ATS analytics and interactive UI; containerized services using Docker.

Technologies / Tools Used : *FastAPI, RAG, PostgreSQL, Redis, JWT, OAuth, React.js, Chart.js, Groq API*

AP & Telangana House Price Prediction | Github Website

- Developed an end-to-end ML prediction system for residential property prices across **61 districts** in Andhra Pradesh and Telangana, using property, location, and amenity features.
- Performed data preprocessing and high-cardinality categorical encoding for **700+** localities, and trained/evaluated regression models using cross-validation.
- Achieved **84.41% R^2**, with **MAE: 12,896**, **MSE: 232.28M**, and **RMSE: 15,240**, demonstrating strong predictive performance.
- Built and deployed a **Dockerized FastAPI REST API** for real-time price predictions, with a React-based frontend deployed separately.

Technologies / Tools Used : *Machine Learing, Fastapi, Docker, React*

SKILLS

Programming Languages :	Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
Frontend Development :	React.js, Responsive Web Design, REST API Integration, Axios
Backend Development :	FastAPI, Django, Django REST Framework, RESTful APIs, SQLAlchemy, Pydantic, JWT Authentication, OAuth 2.0
AIML :	Machine Learning, Scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib
Generative AI & Agentic AI :	LangChain, LangGraph, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, LLM Integration, Tool Calling, Embeddings, Vector Search, Semantic Search
Databases & Vector Stores :	PostgreSQL, Supabase, Firebase, Redis, pgvector, FAISS
Developer Tools & Platforms :	Git, GitHub, Docker, Postman, VS Code, Vercel, Render, Jupyter Notebook
Computer Science Fundamentals :	Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Database Management, Software Engineering
Professional Skills :	Problem Solving, Analytical Thinking, Communication, Teamwork, Leadership

AWARDS & ACHIEVEMENTS

- Secured **3rd Prize** in the Gen-AI Hackathon 2025.
- Achieved **2nd Place** in the Shashtra Coding Competition.
- Successfully solved over **220+ LeetCode problems**.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Data Science Professional – **Oracle**
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional – **Oracle**
- AI-ML Engineer Certificate Course (150 Hours) – **Reliance Foundation**
- Python for Data Science (Elite) – **NPTEL, IIT Madras**
- Fundamentals of Agents – **Hugging Face Agents Course**
- Python Full Stack Internship (10 Weeks) – **EduSkills Academy**
- Python (Basic) Skill Certification – **HackerRank**